

High-voltage Temperature Humidity Bias Test And F-Lab Volta, France

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F-LAB VOLTA, FRANCE

Founded in 1935

Grenoble has 2 facilities:

- Technopole
- Electropole.

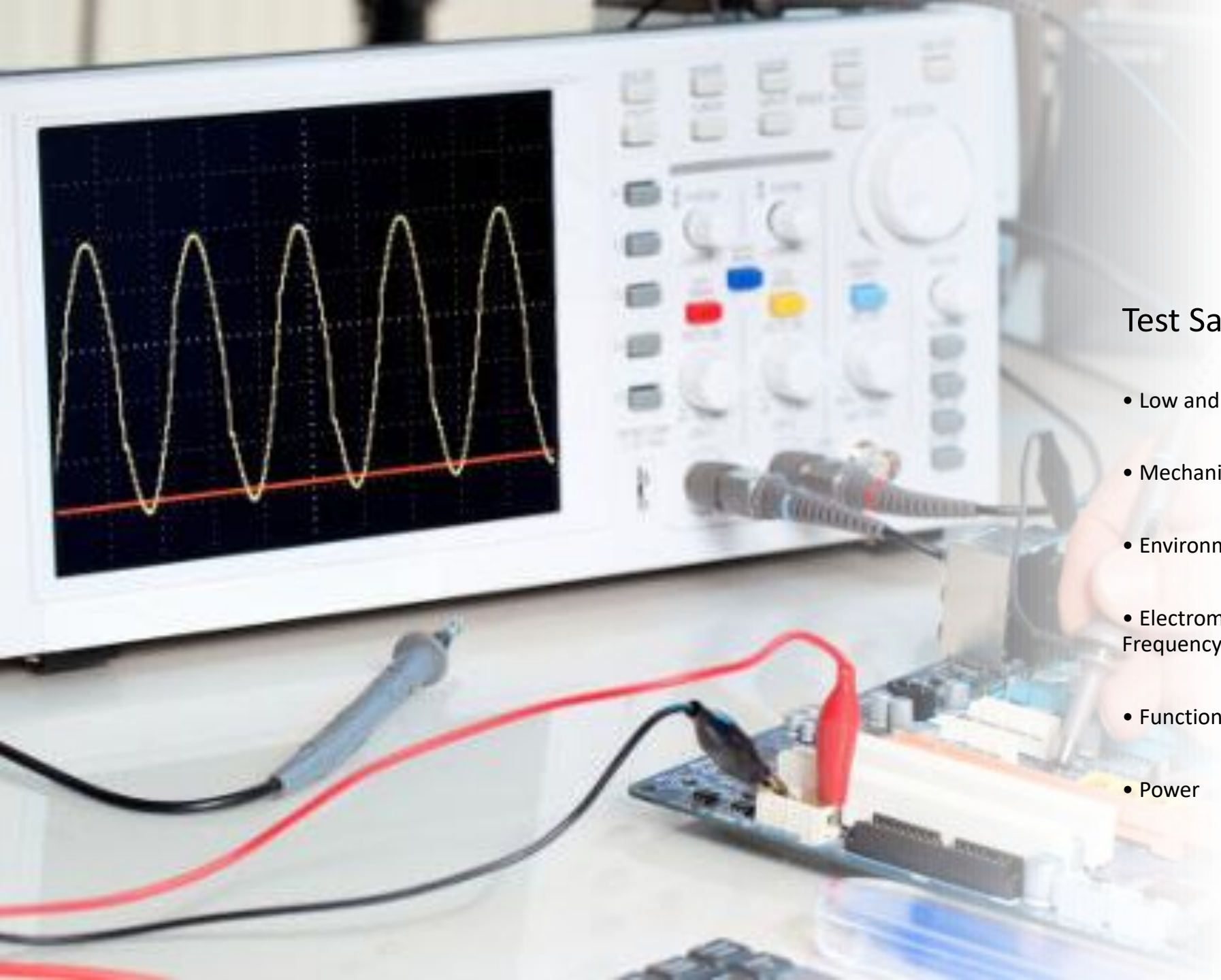
STANDARTS:

- EC
- IEEE
- ANSI
- UL





- Testing and Expertise
- Employee
- Investment
- Experience
- Accreditations



Test Samples in the Lab:

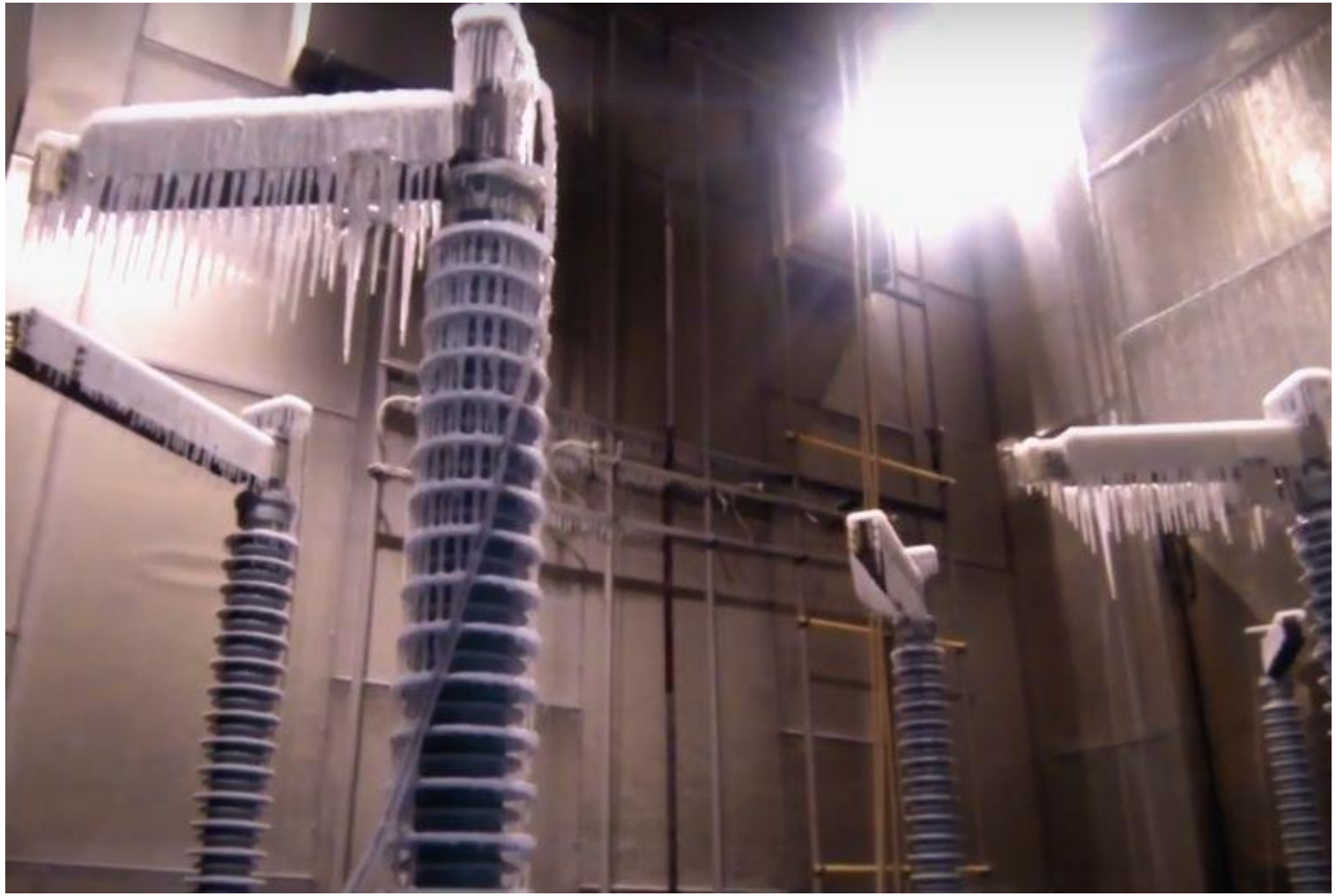
- Low and medium voltage Dielectric & Electrics
- Mechanical
- Environment, Climatic
- Electromagnetic Compatibility (EMC) & Radio Frequency
- Functional
- Power

Functional tests

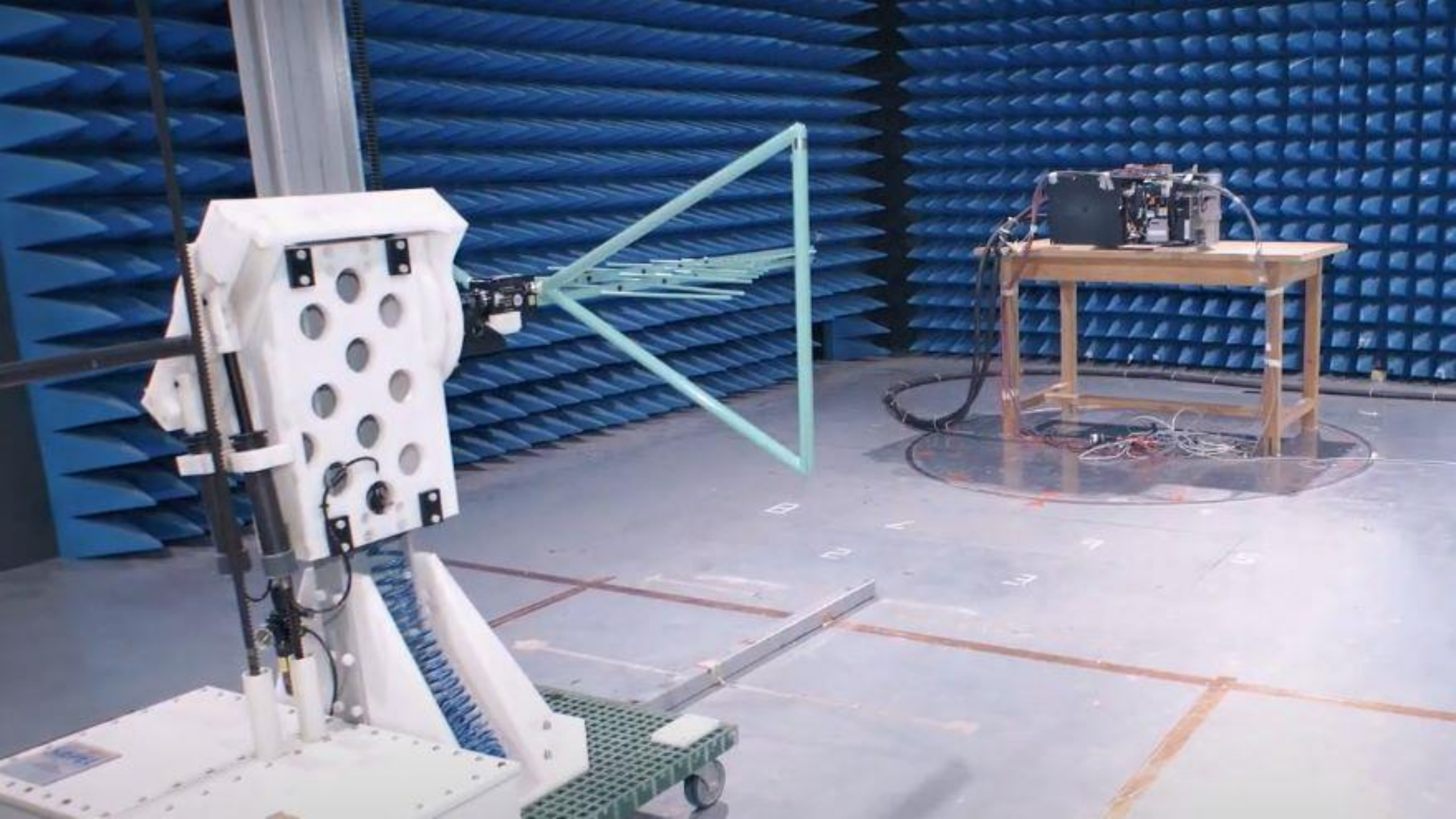
This laboratory proposes a wide range of test services (heating, triggering, dielectric, etc.) at high performance levels and using unique means such as tests on the energy measuring devices. This laboratory mainly conducts the following tests:

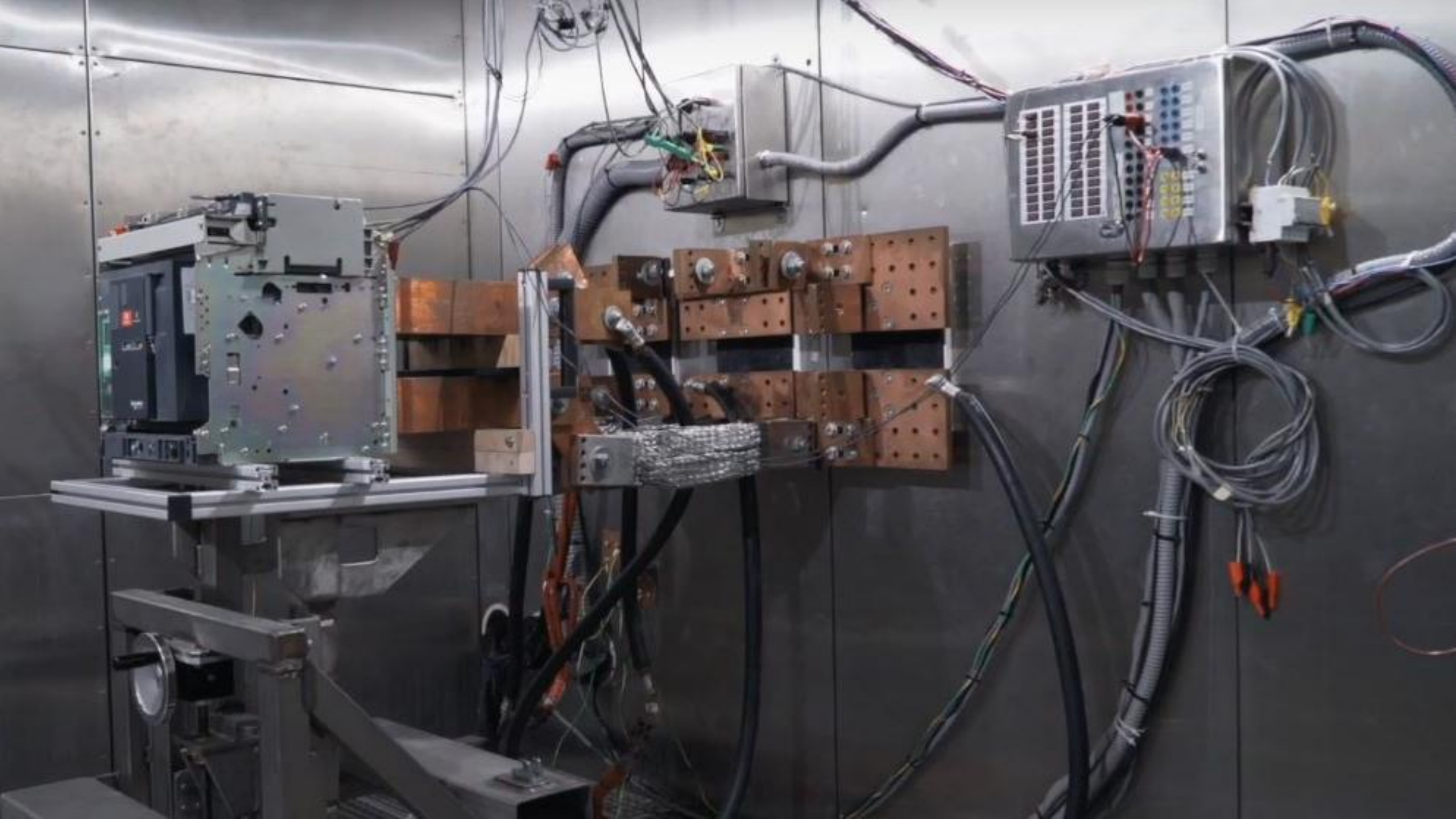
- Heating
- Triggering of overload and short-circuit
- Dielectric (industrial frequency, shock waves and current shocks, etc.)
- Aging tests (28 days)
- Disturbance of the electrical network (voltage, current dips, harmonics, frequency and amplitude variations, power loss, etc.)
- Mechanical stresses (torsion and traction)
- Incandescent wire
- Operation at extreme temperatures






GB






Volta
Schneider Electric


CTS
CLIMATS





The laboratory is equipped with several climatic chambers, AC/DC current and voltage sources as well as combined means, of programmable multi-stress type, to conduct automatic test sequences.

Means	Characteristics
Current sources	Various current sources up to:
	8600 A / 30 V three-phase AC
	15 000 A / 30 V single-phase AC
	7000 A / DC
	20 to 1500 Hz
	Harmonic ranks 3 to 30
Climatic chambers	Climatic chambers from 1 to 100 m^3 (-40 °C to +80 °C)
	230 m^3 climatic chambers (-65 °C to +85 °C; 0 to 100 % RH + water and ice)

Tests	Means	Characteristics
Industrial frequency	Transformer	50 Hz; 2 x 150 kV
Lightning shocks	Marx generator	800 kV; 5 kJ per stage
Operating shocks	Marx generator	up to 600 kV
Pulse current shocks	Current generator	100 kV; 50 kJ: Wave form: 8/20 µs: 60 kA/10 kV Wave form: 4/10 µs: 100 kA/10 kV
Corona and partial discharge tests	Partial discharge detector with a high voltage source	Up to 100 kV

High Voltage Dielectric Tests

- The mission of the high voltage dielectric laboratory is to check product insulation and resistance to external overvoltages.
- The tests are conducted on various types of equipment, such as high voltage cells and switchboards (HVA/HVB), insulating materials, lightning arresters and voltage limiters.
- Types of test conducted in the laboratory :

Name	Type	Volume	Measurement type
3D machine	With probe contact	700 x 1000 x 700 mm	3D
Measuring arm	With probe contact	Volume 1 m^3	3D
Optical machine	Optical without contact	400x350x300 mm	2D/3D
Optical machine	Optical without contact	300x200x200mm	2D/3D
Scanner measuring arm	Laser without contact	Volume 1 m^3	3D digitization of the part
CT scanner	X-ray without contact	Cylinder (d)120x(h)250 mm	3D digitization of the part
Confocal miscroscope	Without contact	120x200x150 mm	2D/3D

Mechanical measurements

- The laboratory is equipped with measuring equipment covering all the mechanical functions of the products

LOW VOLTAGE TESTS

Low voltage tests – Voltage Current		
	U Max	I Max
AC	2200 V	300 kA / 1 s (700 kA peak)
DC	2000 V	200 kA / 1 s

- The bench used to conduct short-circuit make and break tests checks the ability of the device to protect the installation and the persons if a short circuit occurs on the network:

Climatic and Vibration Laboratory

Sine Wave/Random vibration and Shock bench

The sine wave or random vibration and shock bench is used to check that the product operates correctly under severe mechanical stresses with the use of vibrating pots. It subjects the product to a mechanical environment with the following characteristics:

- A nominal force in permanent sinusoidal regime at 57 kN
- A nominal peak-to-peak displacement of 63.5 mm



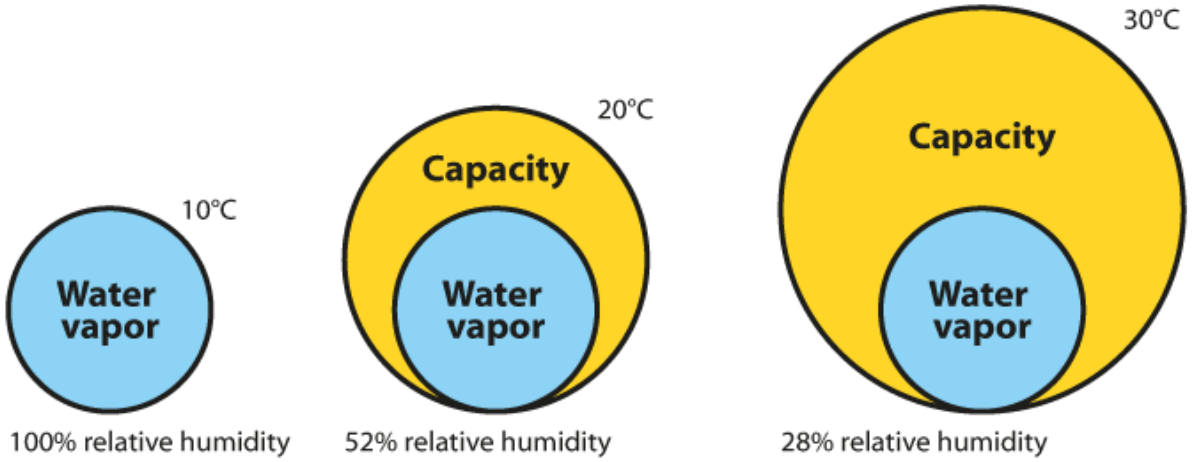
Stock of Climatic Chambers

The stock of climatic chambers is used to check that the product operates correctly in a severe climatic environment. Chambers of volumes of up to 15 m3 are available. The product is subjected to various climatic conditions such as :

- Temperatures from -70 °C to +180 °C
- Humidity up to 98 % RH
- Rapid temperature variation: 15 °C/min
- Thermal shock from -55 °C to +180 °C



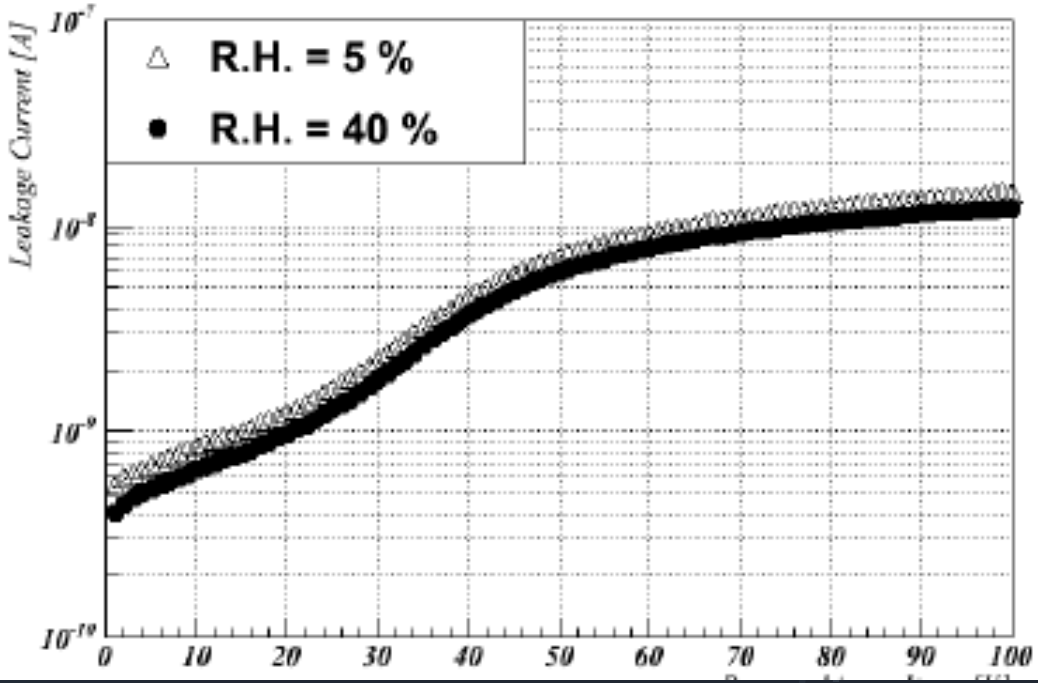
TEMPERATURE HUMIDITY TEST

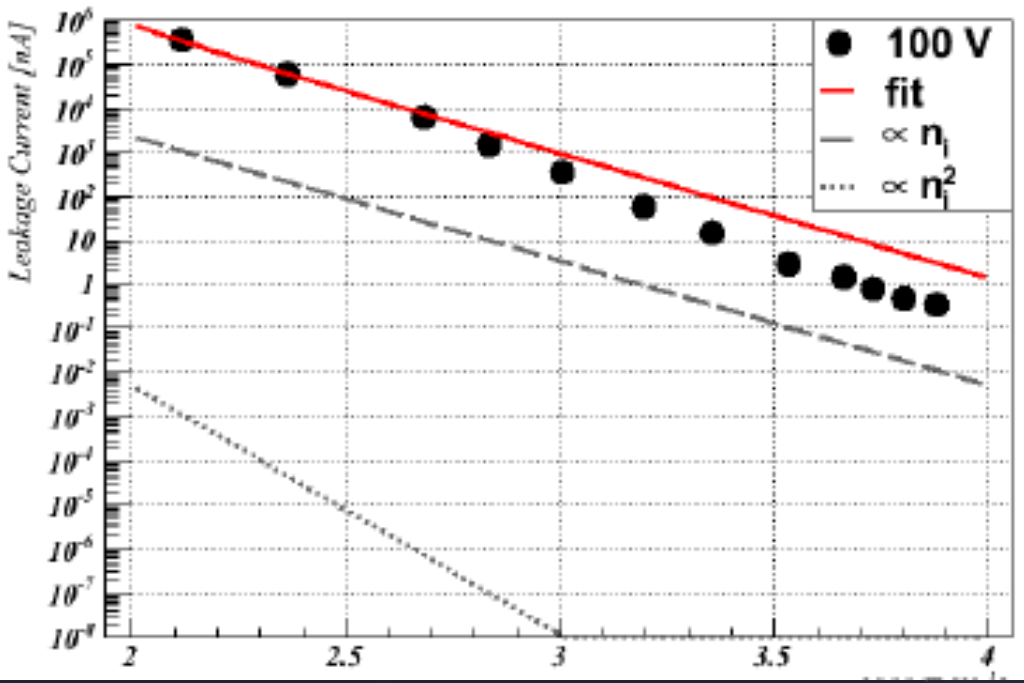






- Semiconductors





- The fit shows that the leakage current is dominated by the generation current.

2. Test Methodologies

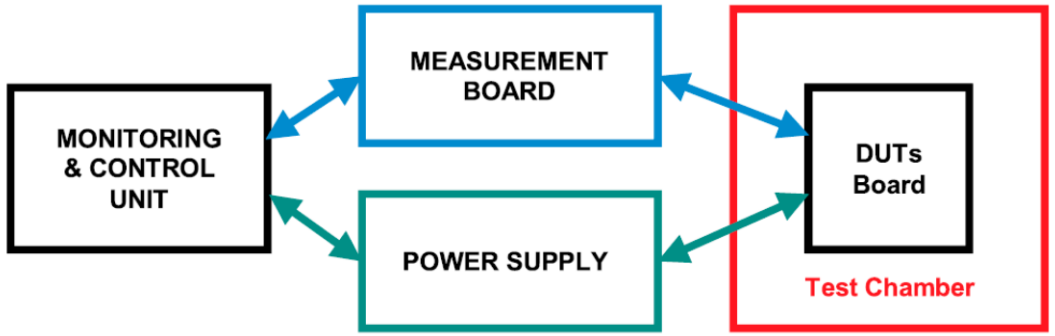
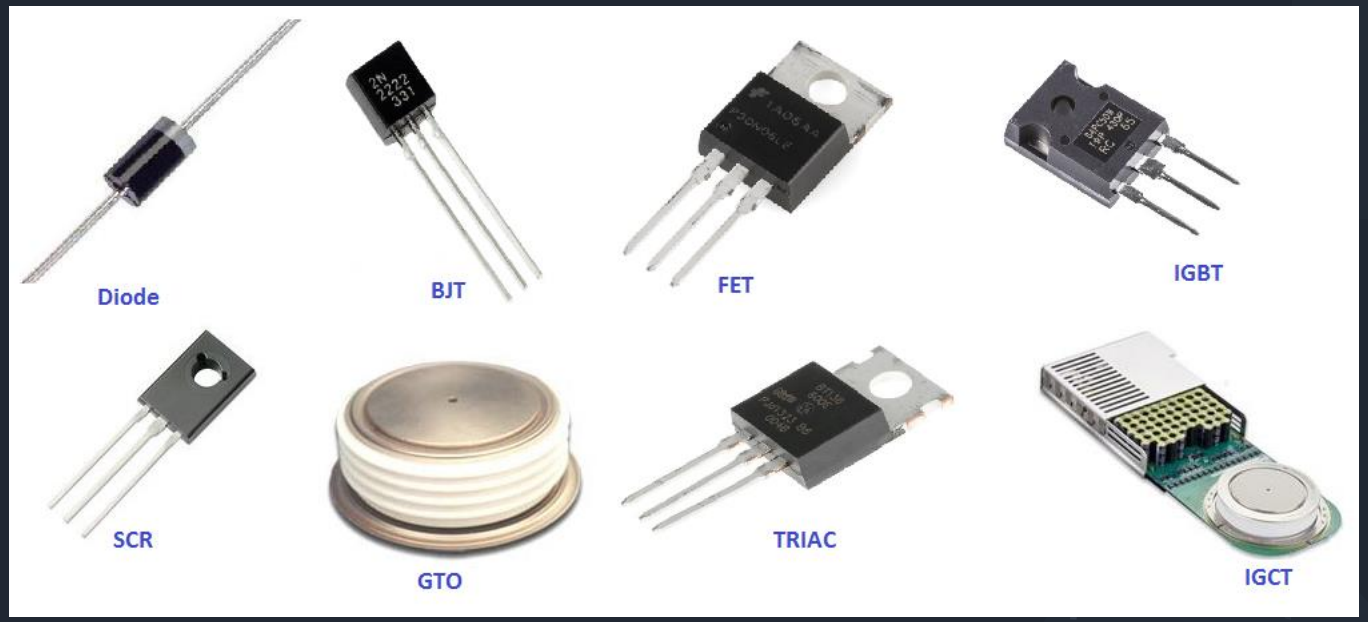


Figure 2.1 Diagram of Test Setup



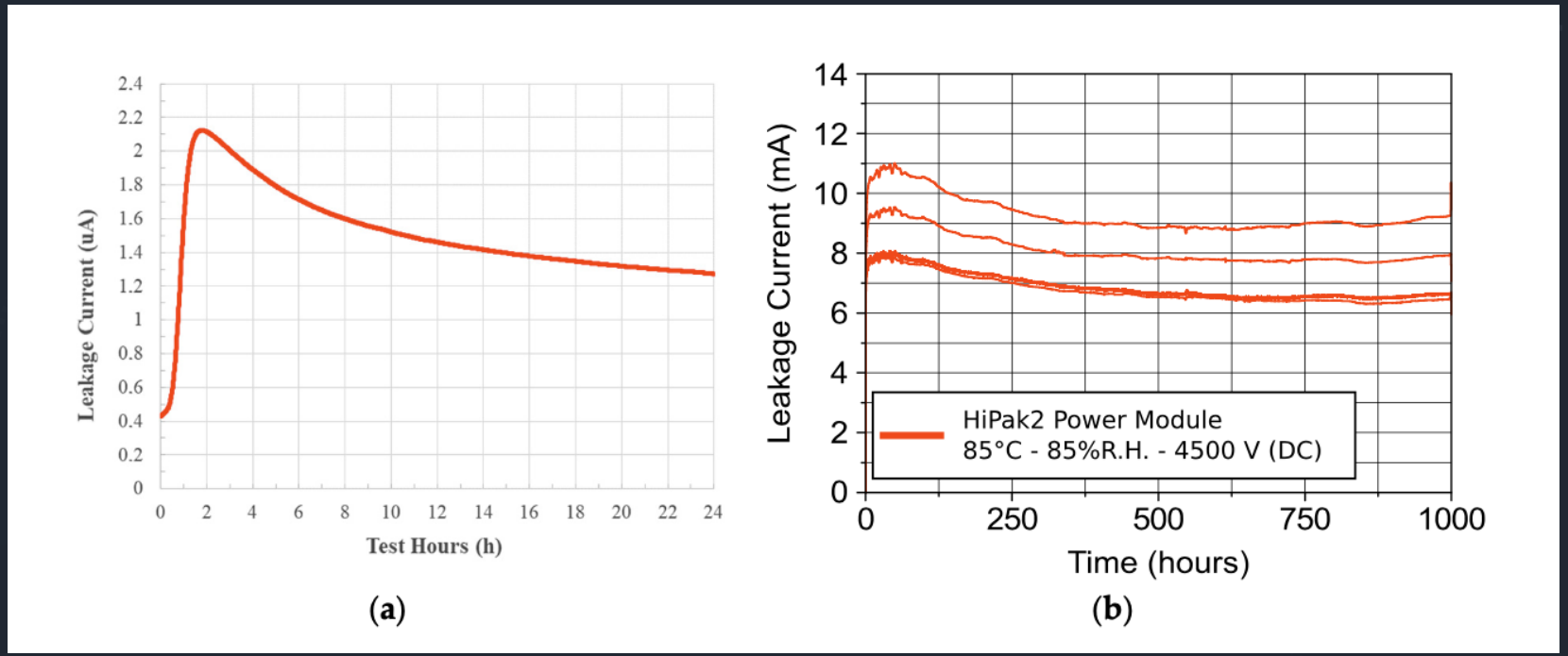

Diode
BJT
FET
IGBT
SCR
GTO
TRIAC
IGCT

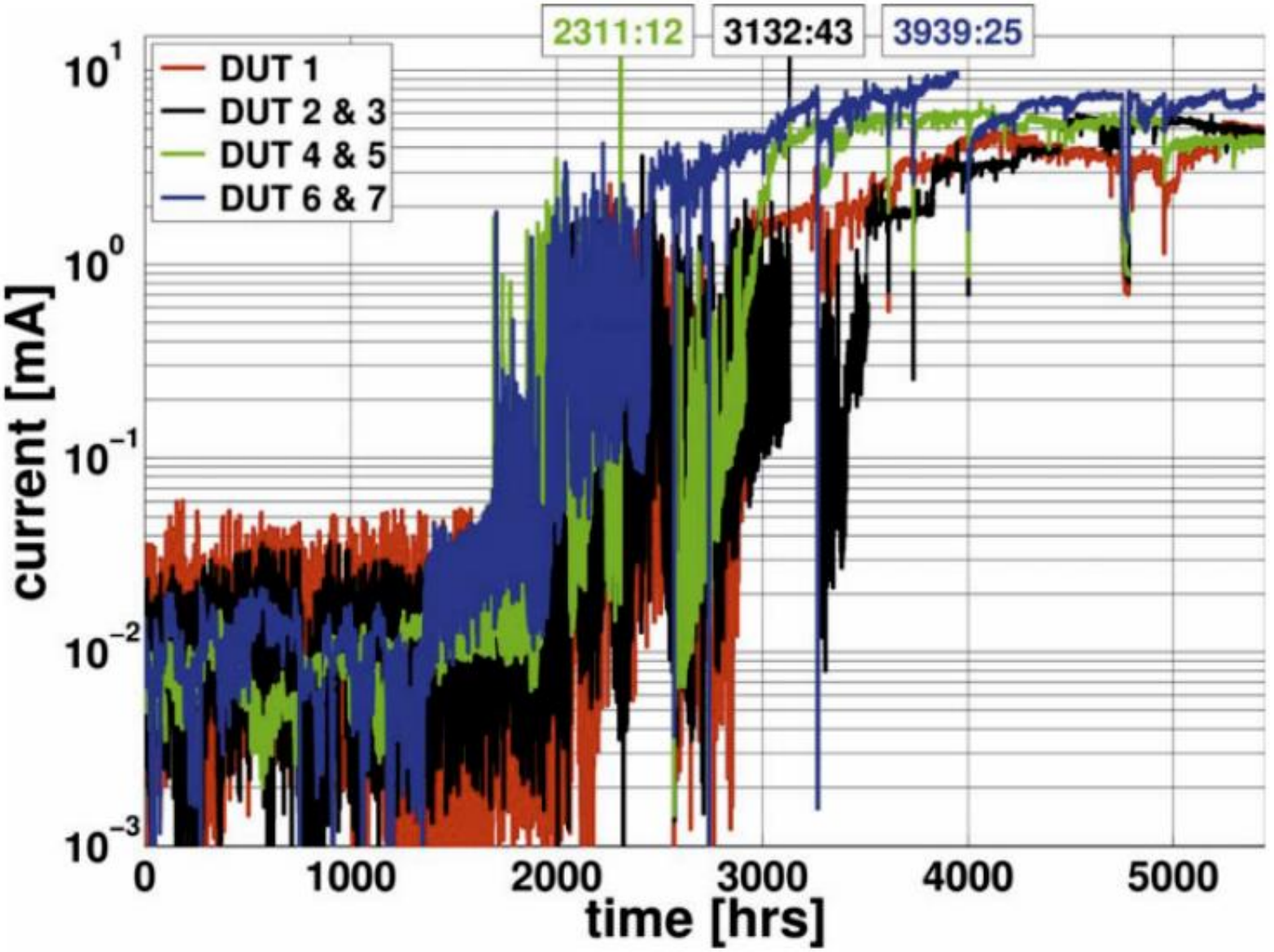
2.1. Testing Procedure

1. Observe reverse I-V
2. Apply reverse voltage
3. Wait
4. Repeat

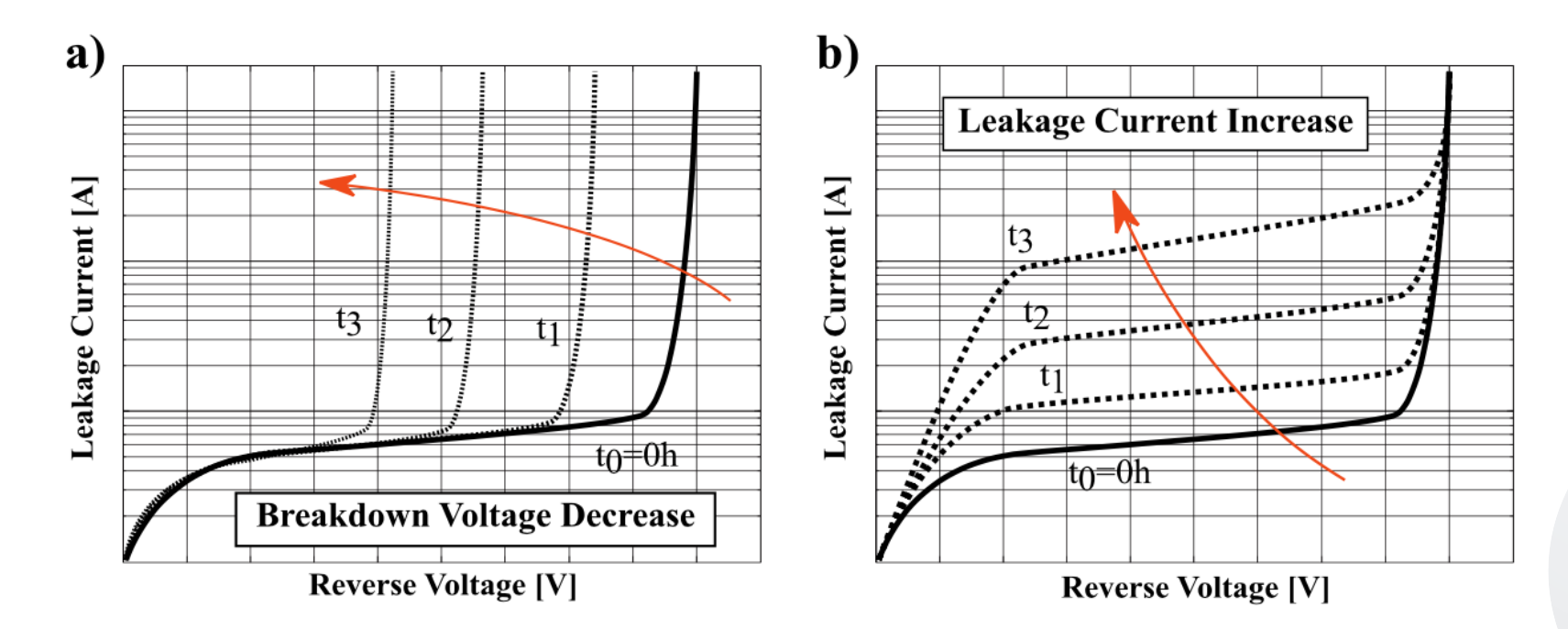


2.2. Leakage Current Monitoring

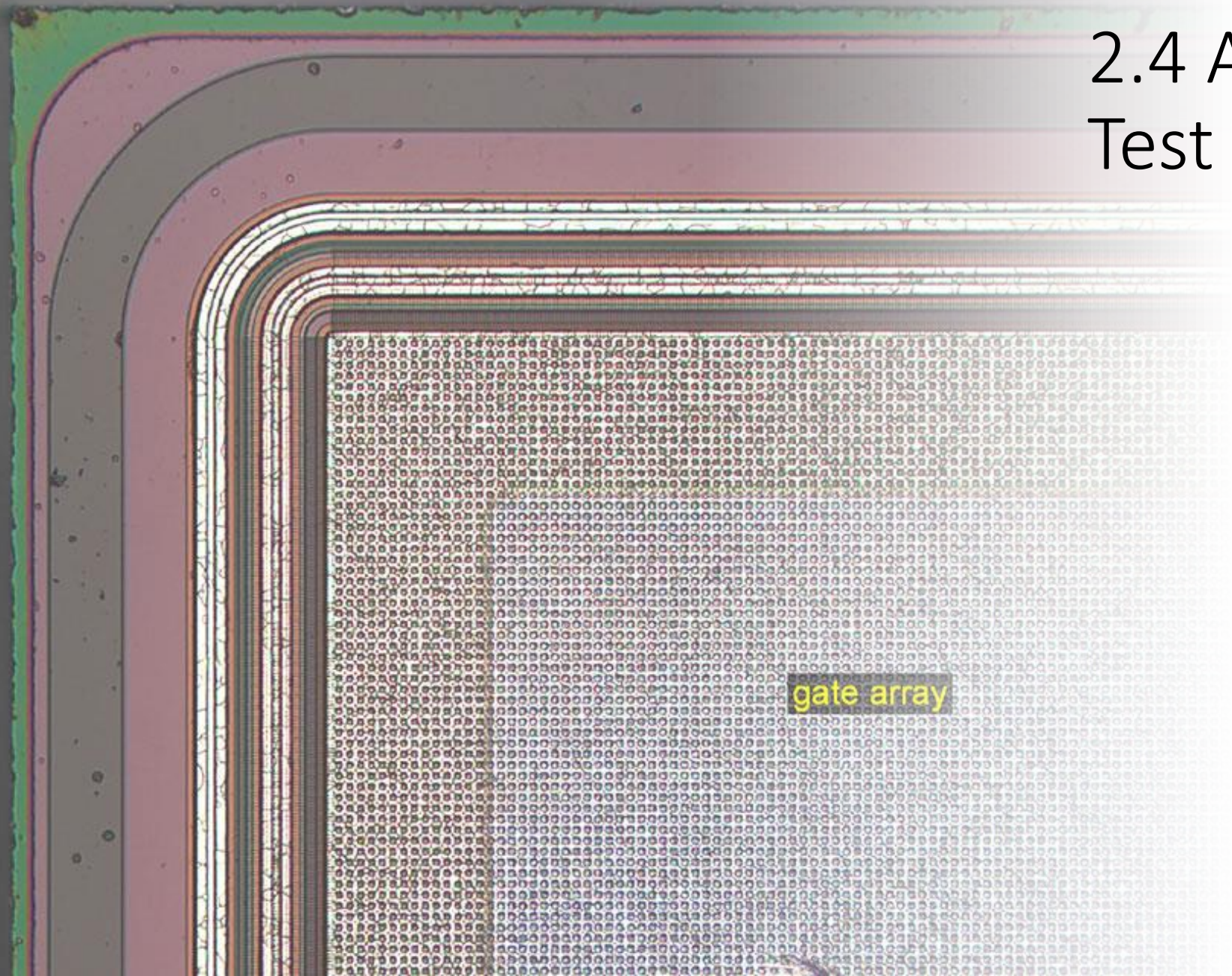



2311:12
3132:43
3939:25
DUT 1
DUT 2 & 3
DUT 4 & 5
DUT 6 & 7
current [mA]
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
1000
2000
3000
4000
5000
time [hrs]

Electrical Degragation



2.4 Accelerated Test Models

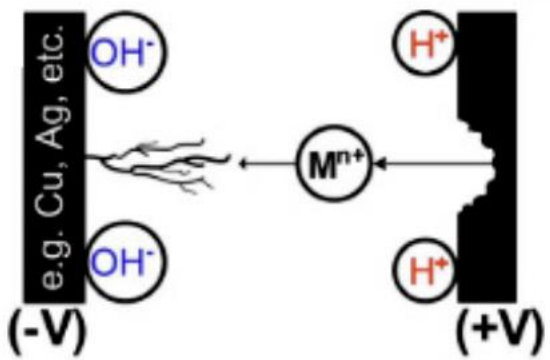


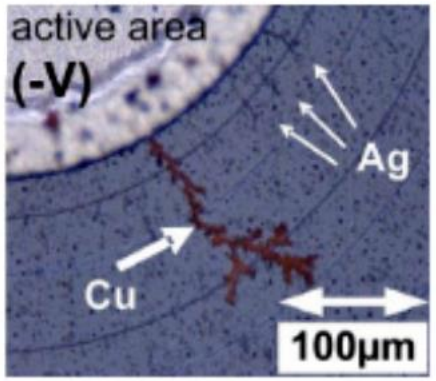
$$t_f = \frac{Q_c}{\beta \cdot J_{tip}} \quad (1)$$

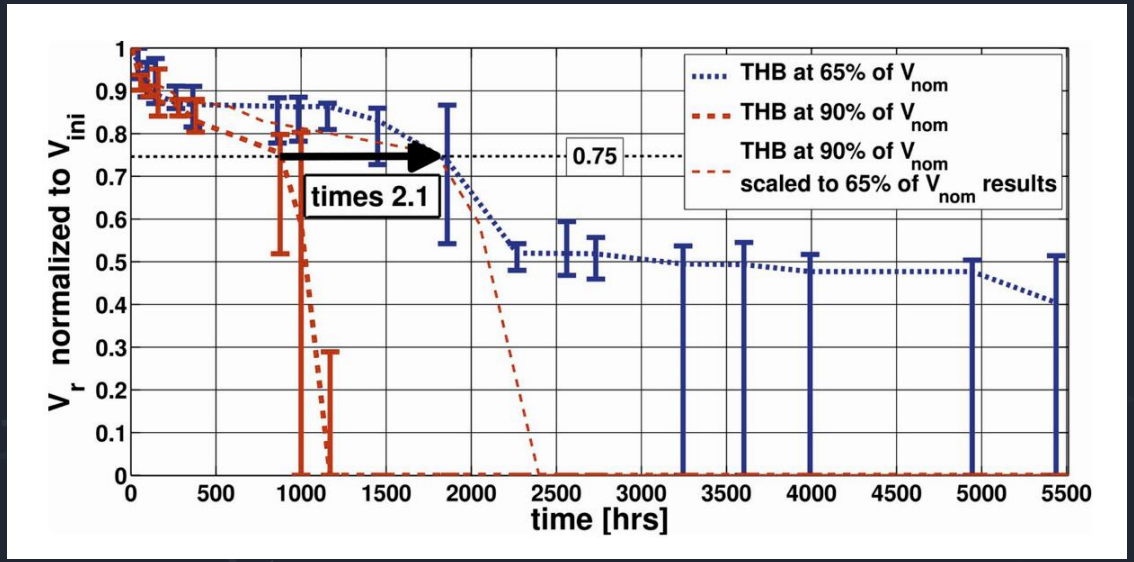
$$t_f = A_1 \cdot \exp\left(\frac{E_A}{k \cdot T}\right) \quad (2)$$

$$t_f = \frac{\alpha \cdot d}{V} \cdot \exp\left(\frac{E_A}{k \cdot T}\right) \quad (3)$$

$$t_f = A_3 \cdot RH^{-x} \exp\left(\frac{E_A}{k \cdot T}\right) \quad (4)$$



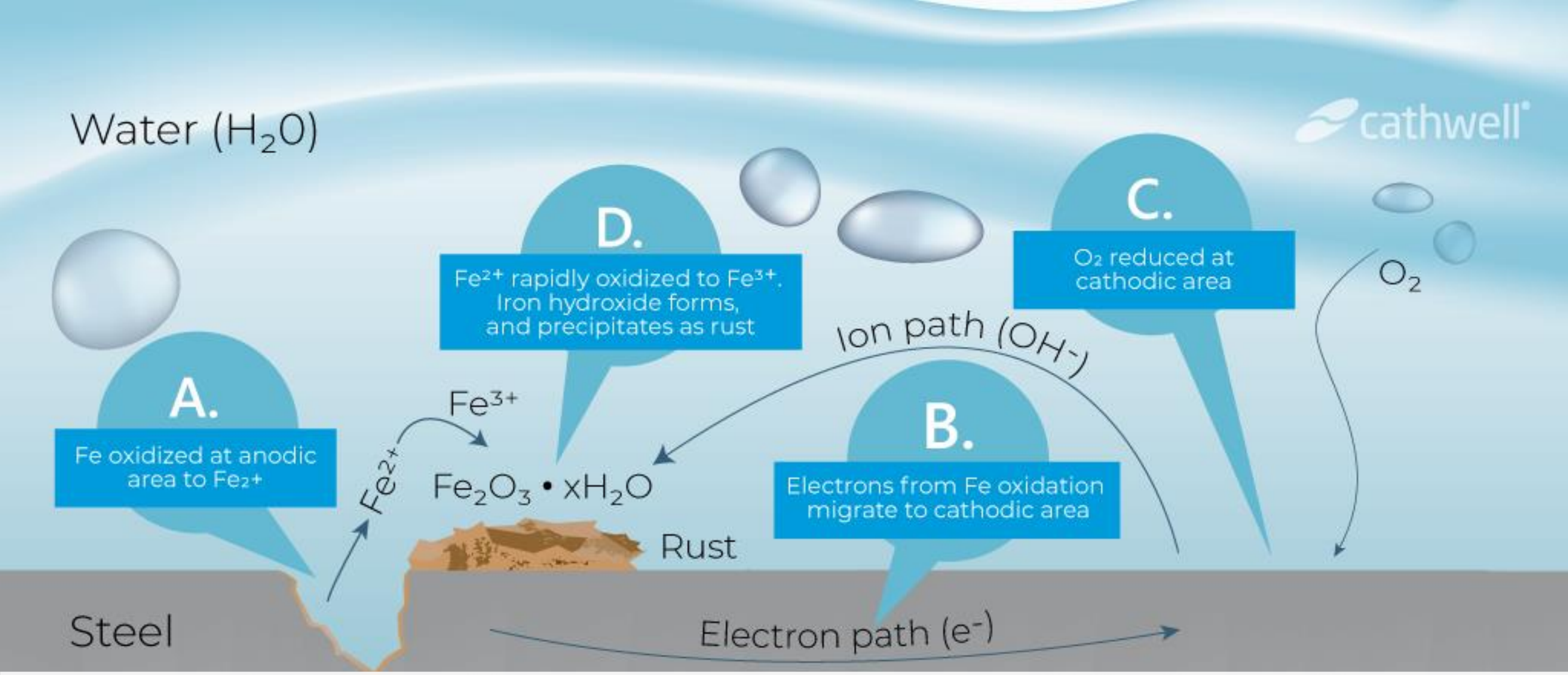




$$a_f(RH, T, V) = \left(\frac{RH_a}{RH_u}\right)^x \cdot \exp\left(\frac{E_A}{k} \cdot \left[\frac{1}{T_u} - \frac{1}{T_a}\right]\right) \cdot \left(\frac{V_a}{V_u}\right)^y \quad (5)$$



3. An Overview of the Failure Modes



3.1. The Corrosion Cell

Anodic area (+)

Where metal dissol

Cathodic area (-)

Electron conduction

$Fe \rightarrow Fe^{2+}+2e$

$O_2+2H_2O+4e^- \rightarrow 4OH^-$

$$2\,H_2O \rightarrow O_{2(g)} + 4H^+ + 4\,e^-$$

$$2H_2O + 2e^- \rightarrow H_{2(g)} + 2(OH)^-$$

$$2H_2O + O_2 + 4e^- \rightarrow 4(OH)^-$$



3.2. Aluminum Corrosion

$$Al(OH)_3 + 3H \leftrightharpoons Al^{3+} + 3H_2O$$

$$Al(OH)_3 + OH^- \leftrightharpoons [Al(OH)_3]^-$$

$$Al + 3H_2O \rightarrow Al(OH)_3 + \frac{3}{2}H_2$$

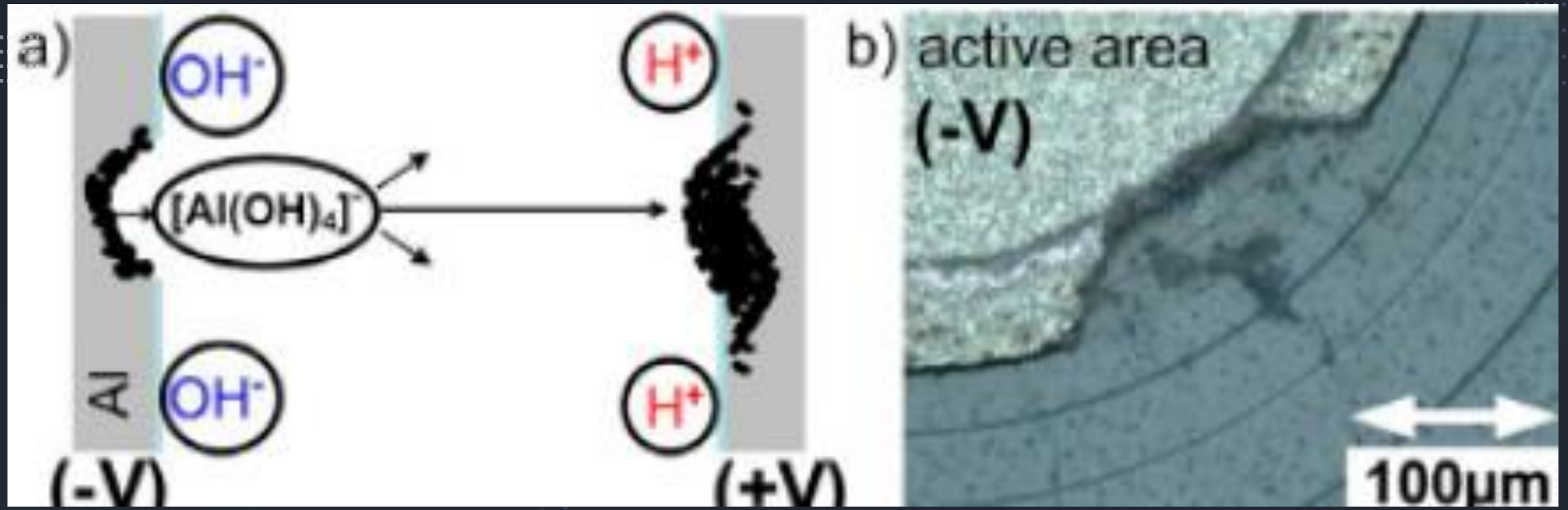


Figure A

- Figure B

3.3. Electrochemical Migration (ECM)

$$M \rightarrow M^{n+} + ne^{-}$$

$$M^{n+} + ne^{-} \rightarrow M$$

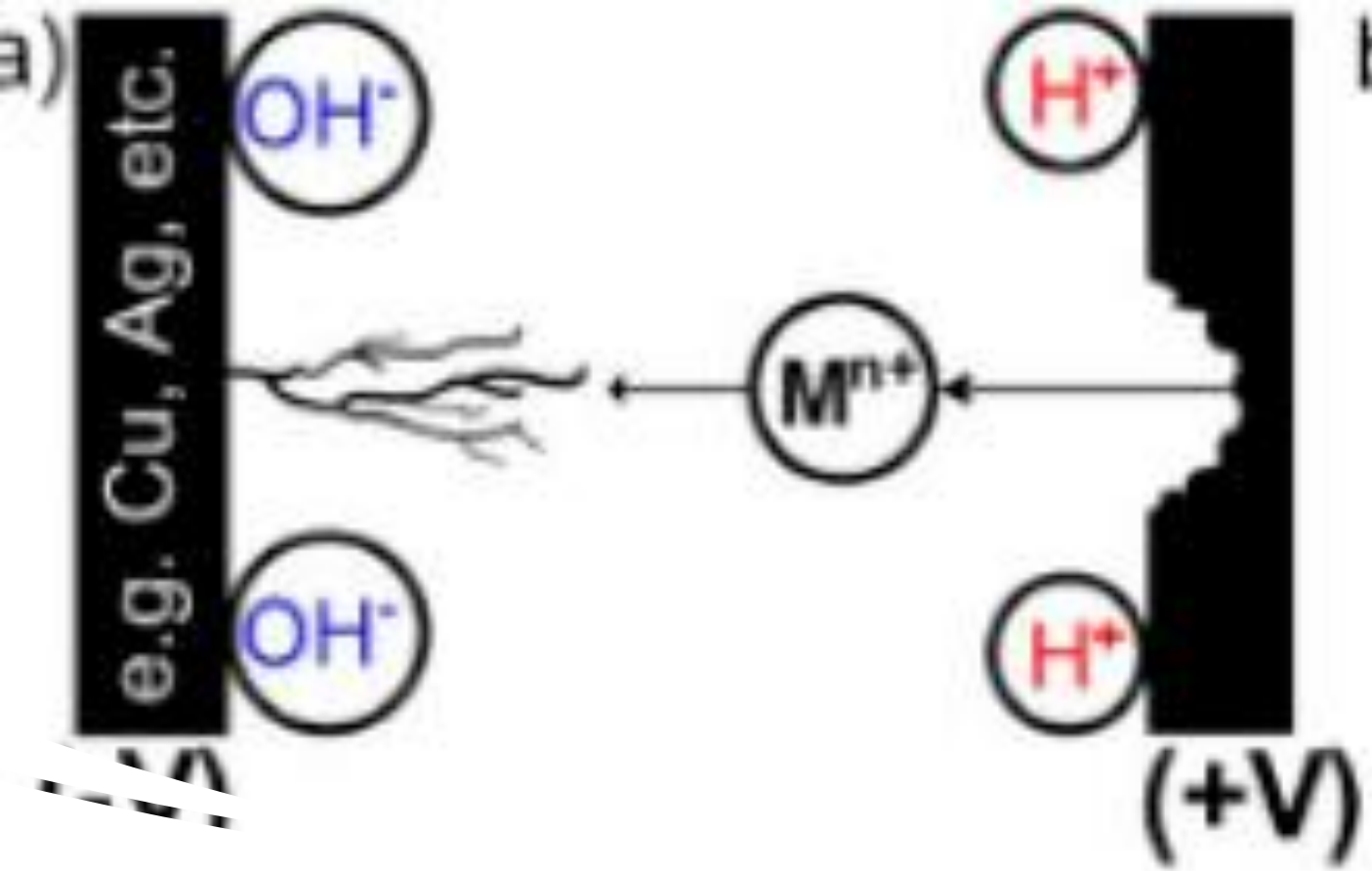


Figure A

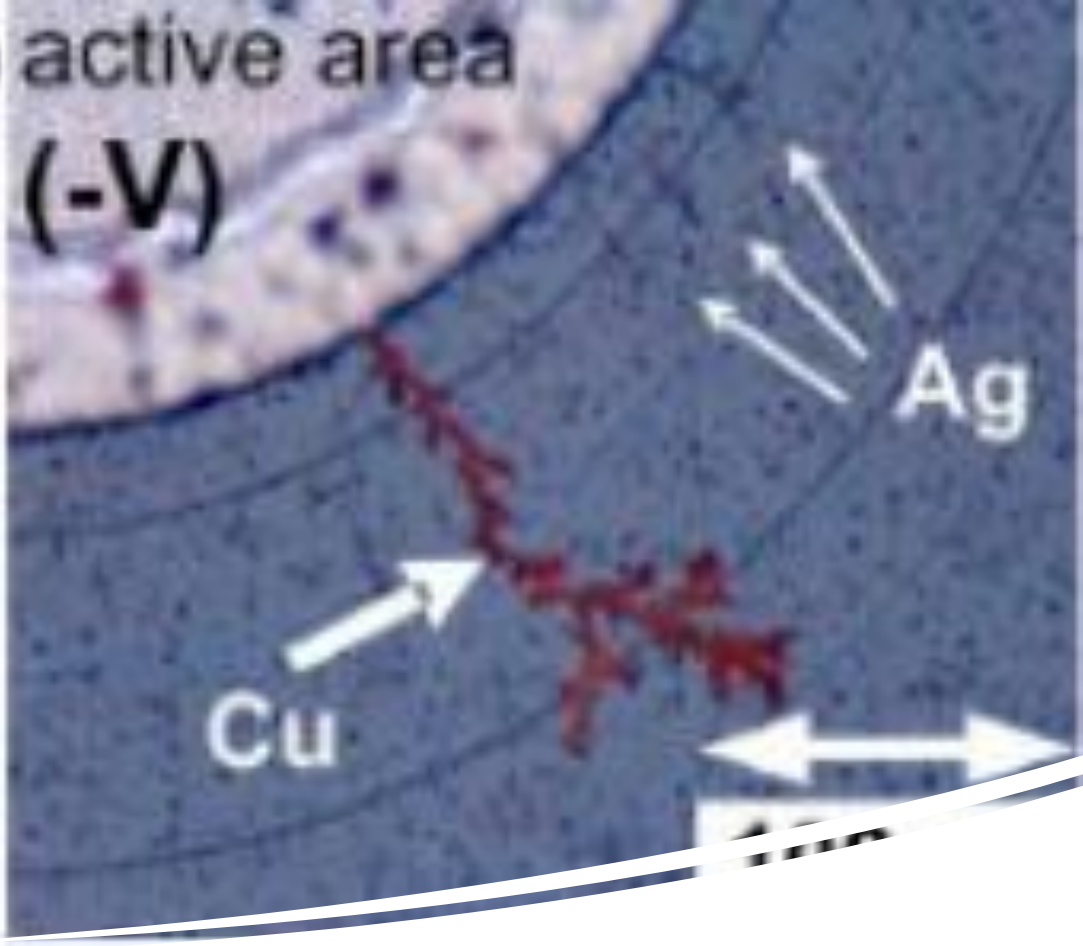
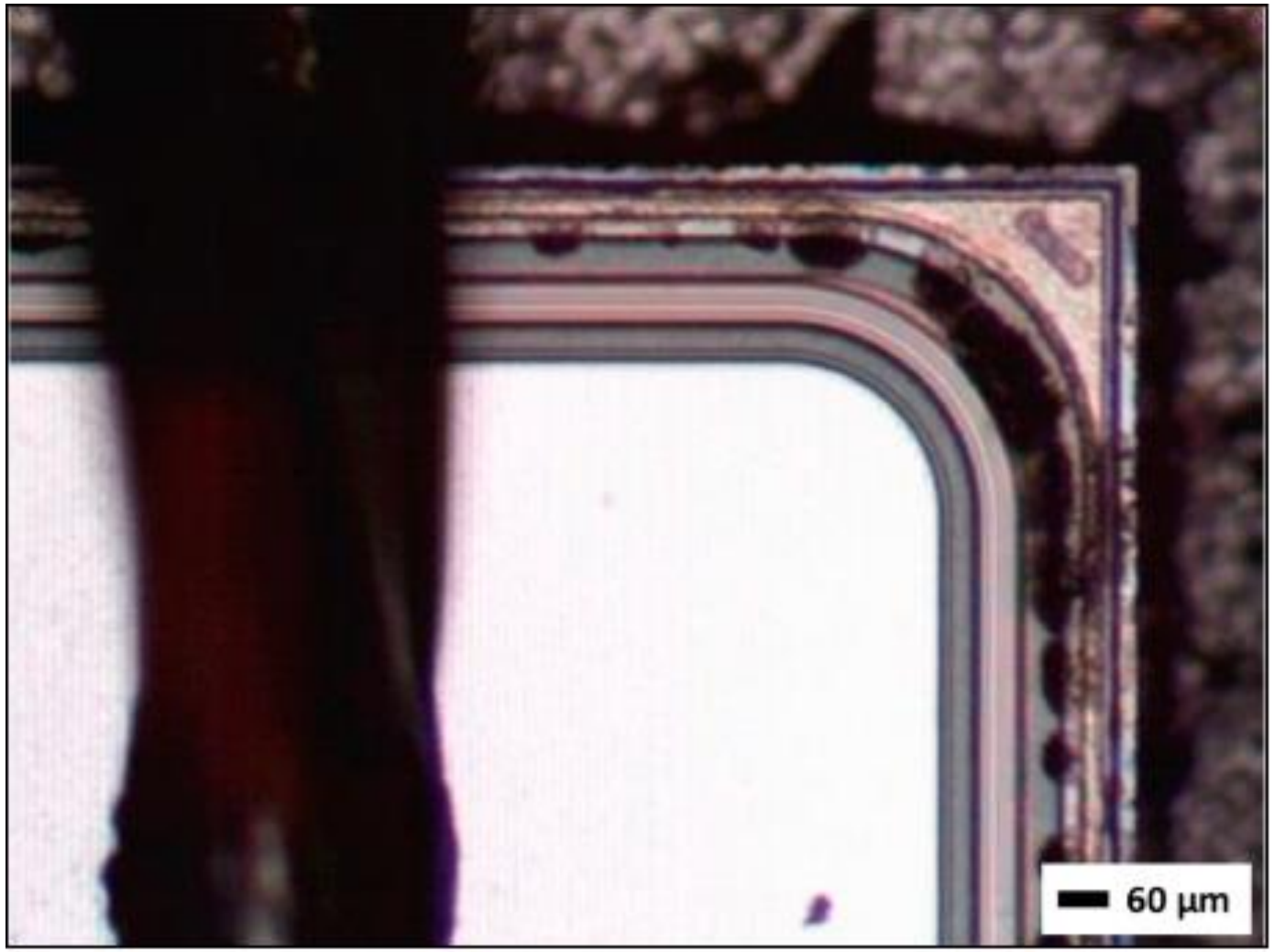
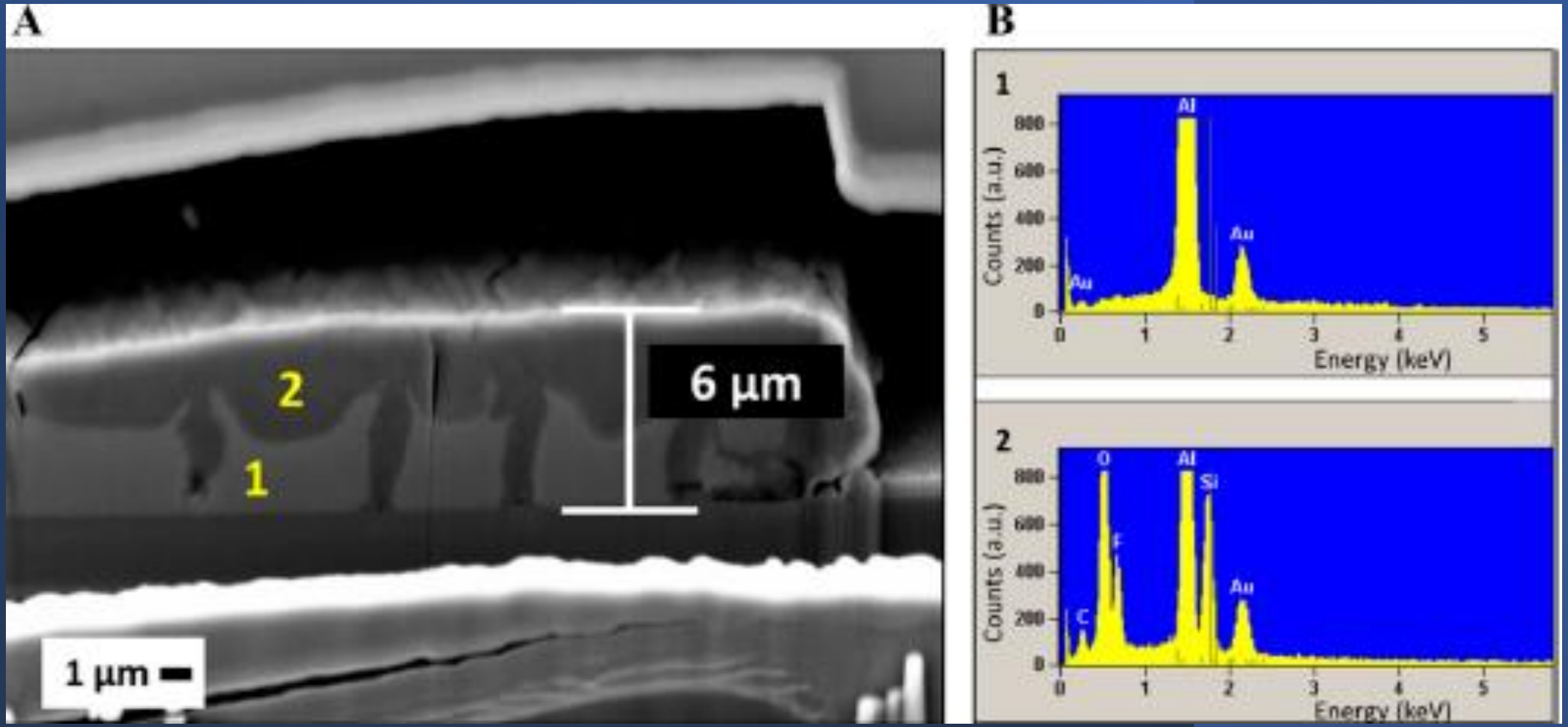


Figure B

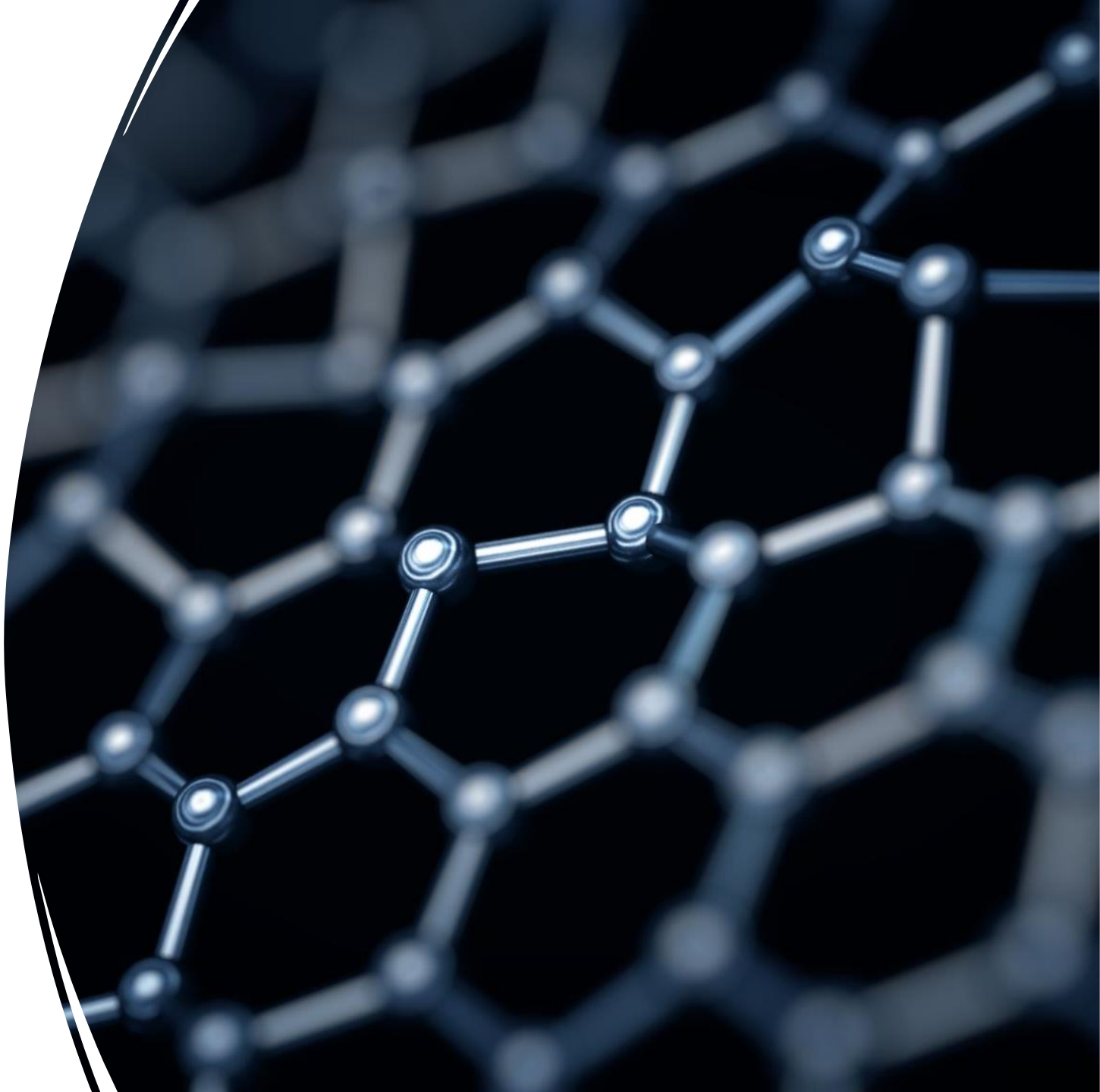
3.4. Device Failure Analysis




A
2
1
6 µm
1 µm
B
1
Al
Au
Au
Counts (a.u.)
Energy (keV)
2
O
Al
Si
F
C
Au
Counts (a.u.)
Energy (keV)

4. Materials and Accelerated Testing Performances

- 4.1. Passivation and Termination Materials
- 4.2. Silicon Nitride as a Passivation Material
- 4.3. Silicon Carbide Devices



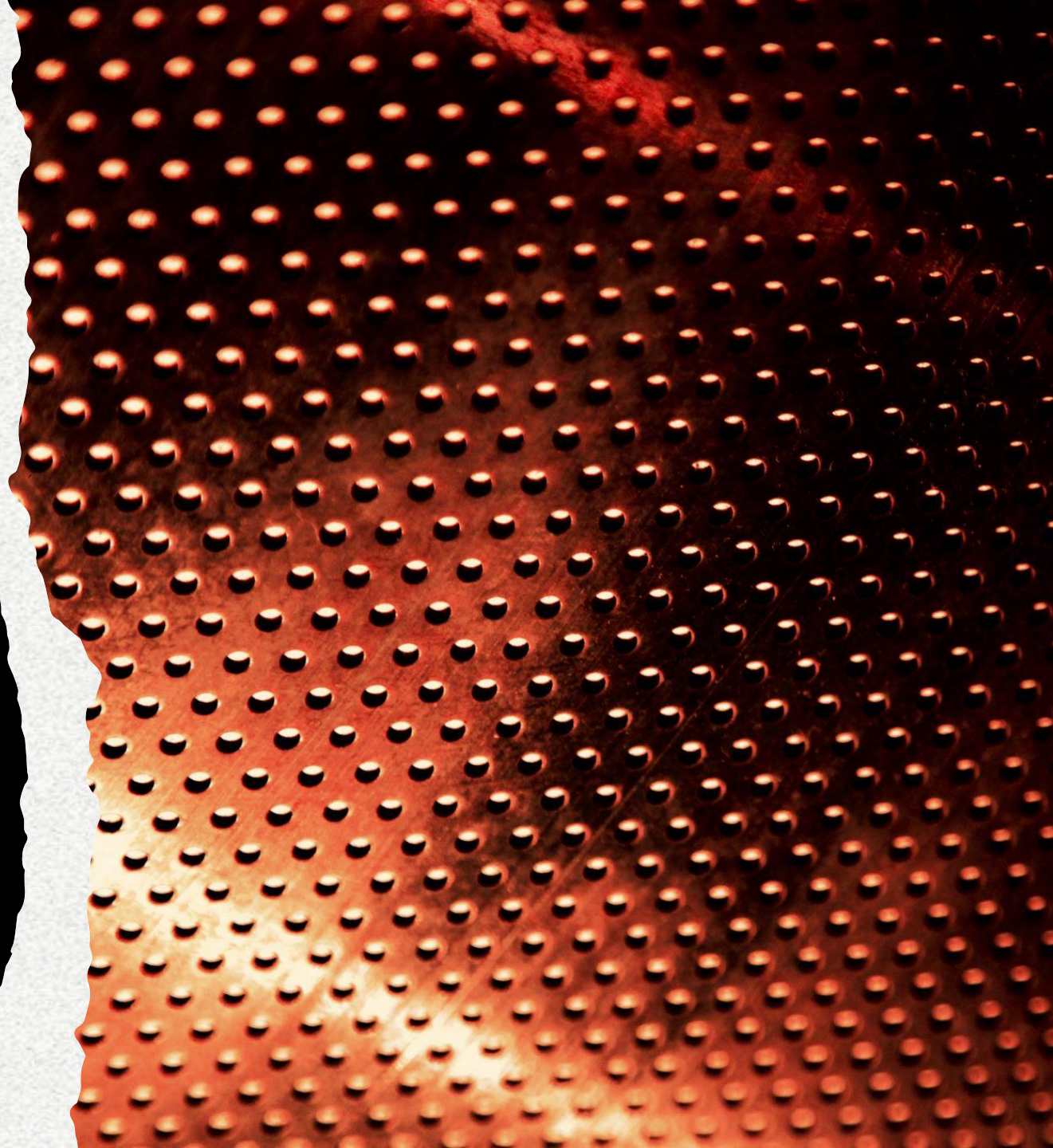
Passivation and Termination Materials

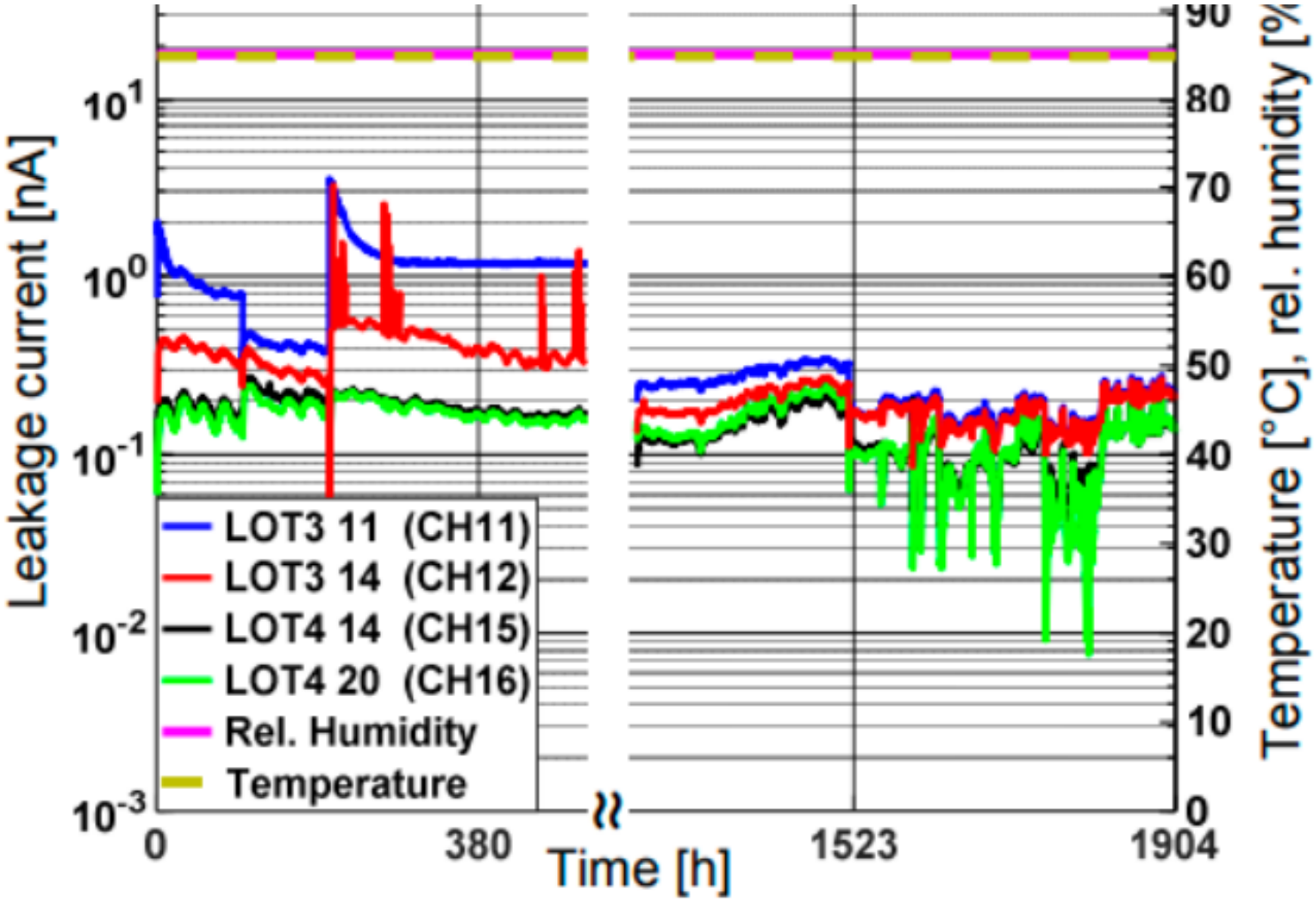
Material	Relative Dielectric Constant	Band-Gap (eV)	Critical Electric Field (MV/cm)	Thermal Conductivity (W/cmK)
Al_2O_3	8	8.8	>5	0.02 #
AlN	9.14	6.03	1.2 ÷ 1.8	11.7
CaF_2	6.81	12.3	14.44	0.1
HfO_2	~30	6	8.5	0.015 #
LiF	9	11.6	12.24	0.15 ##
Si_3N_4	7.4	5.3	10	0.3
SiO_2	3.9	9	10	0.015 #
TiO_2	24 ÷ 57	3.05	2.7	0.07 #
ZrO_2	15	5.8	15 ÷ 20	0.02
Si	11.7	1.12	0.3	1.5
4H-SiC	9.66	3.23	3 ÷ 5	3.7

Thermal conductivity data for sputtered material; otherwise for bulk; ## thermal conductivity at 77K.

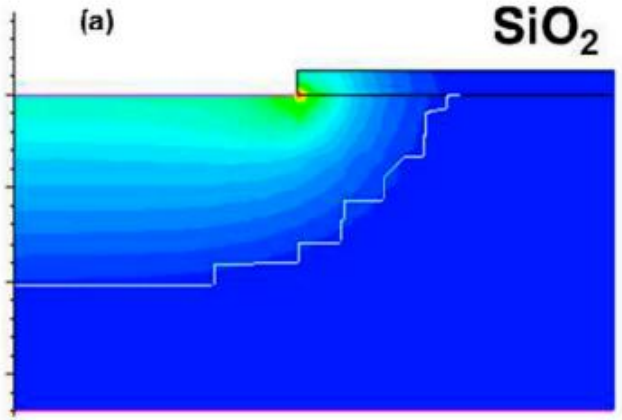
Silicon Nitride as a Passivation Material

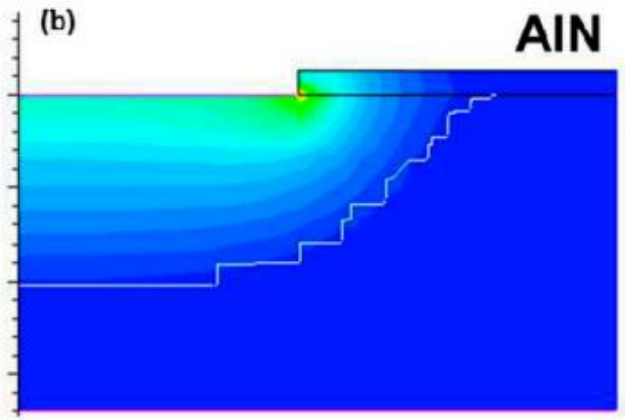
- Silicon nitride Si3N4 is generally involved in the fabrication of high-reliable devices, due to its multiple advantages. In fact, silicon nitride can act both as a getter and alkali barrier, and at the same time have great resistance with respect to humidity.

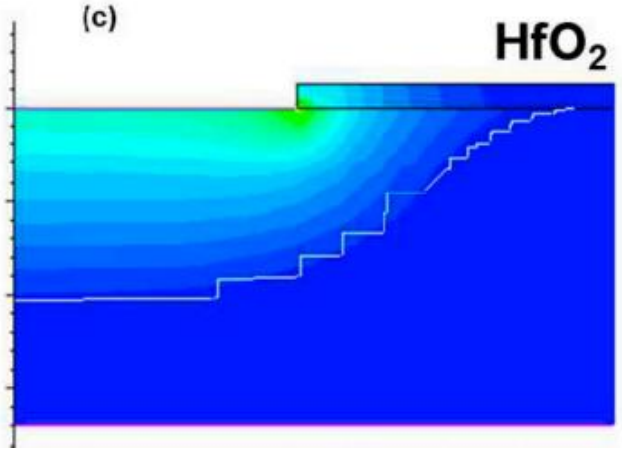


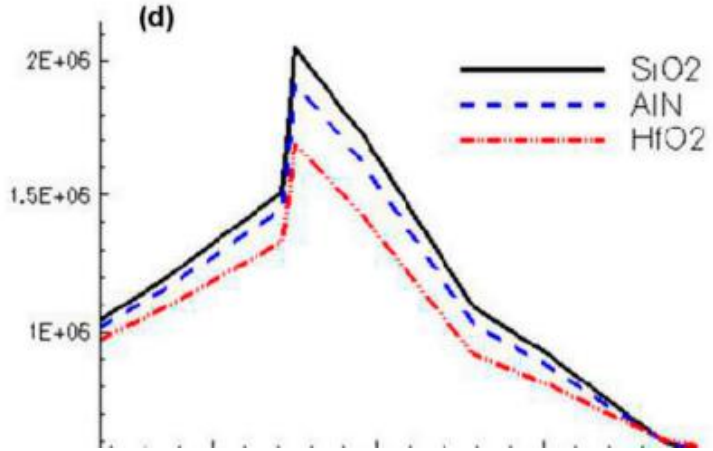


Silicon Carbide Devices


(a)
SiO_2


(b)
AlN


(c)
HfO_2


(d)
2E+06
1.5E+06
1E+06
SiO2
AlN
HfO2

CONCLUSION

- The literature shows that HV-THB testing is a valuable source of information for the evaluation of reliability performances of power semiconductor devices when high bias and humidity are applied simultaneously at a fixed temperature. Moreover, HV-THB will remain a remarkable reliability test, even for new generations of wide band-gap power semiconductor devices.



Contributions

- Mustafa Furkan Eryılmaz 150716016 (Volta LAB introduction and Introduction of the HV Test Part 1)
- Enes Atan 150717001 (Part 2)
- Emre Arat 150719012(Part 3 and Design)
- Yağız Pakçe 150719023 (Tests Conducted in the Volta LAB)
- Fatih Tan 150719684 (Part 4 and Conclusion)



References

- https://www.youtube.com/watch?v=yE4B5dqOVIg
- https://www.espec.co.jp/english/products/measure-semicon/thb/
- High Voltage Temperature Humidity Bias Test (THB) customized system and methodologies for reliability assessment of power semiconductor devices(D. Cimminoa,b,∗, R. Buscaa,b, S. Ferreroa, F. Pirria, G. Richierib, R. Cartab)
- F-Lab-Documentation-compressed(A team of expexrts at your side) by Schneider Electric
- High-Voltage Temperature Humidity Bias Test (HV-THB): Overview of Current Test Methodologies and Reliability Performances Davide Cimmino * and Sergio Ferrero
- The influence of humidity on the high voltage blocking reliability of power IGBT modules and means of protection Charalampos Papadopoulos∗, Chiara Corvasce, Arnost Kopta, Daniel Schneider, Gontran Pâques, Munaf Rahimo